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Structured information extraction from geoscientific literature with GeoExtractor

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Abstract

Earth science research increasingly depends on the integration of large volumes of heterogeneous data distributed across publications. A substantial amount of quantitative information remains embedded in unstructured literature, making large-scale data compilation labor-intensive and difficult to maintain. Here, we introduce GeoExtractor, a large language model (LLM)-powered multi-agent framework designed to automatically extract structured information from geoscience literature. GeoExtractor employs a hierarchical extraction strategy that combines an iterative decision–retrieval–generation workflow with a two-stage retrieval mechanism, enabling the recovery of complex multi-instance scientific records from heterogeneous document sources. We evaluate the framework on four benchmark datasets spanning diverse geoscience sub-disciplines, using three LLMs and repeated runs to quantify variability. GeoExtractor generally performs better than the comparison methods on multi-record and multi-field tasks, while its limitations on single-record datasets are also characterized. To demonstrate its practical utility, we compiled a literature-derived database containing 2,259 zircon analyses from 179 publications of the Central Asian Orogenic Belt (CAOB), and the resulting dataset reproduces previously recognized tectonic patterns in the region. These results demonstrate that LLM-based multi-agent systems can facilitate the transformation of unstructured geoscientific literature into structured knowledge resources, substantially reducing the effort required for database construction and accelerating data-driven research across Earth science disciplines.

Introduction

Earth science research generates large volumes of quantitative information describing the composition, age, properties, and evolution of the Earth system. Although much of this information is reported in published articles, a substantial proportion remains embedded within unstructured text, tables, figures, and supplementary materials. Consequently, the reuse of published data often depends on labor-intensive manual compilation, limiting the efficiency of large-scale synthesis and comparative analyses across studies[1]. The rapid growth of geoscientific literature has further widened the gap between data production and data integration, creating a major challenge for emerging data-intensive approaches in Earth system science[2].

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To improve data accessibility, several community database initiatives have been developed for specific geoscience domains, including PetDB[3], GEOROC[4,5], EarthChem[6]. However, they rely heavily on manual curation, cover only a fraction of the published literature, and often operate within disciplinary silos with limited interoperability[7,8]. Moreover, most existing databases are restricted to relatively homogeneous data types within individual sub-disciplines[9]. By contrast, cross-disciplinary quantitative knowledge remains overwhelmingly dispersed across unstructured literature, lacking unified data schemas and semantic standards, and is consequently difficult to systematically integrate and reuse[10]. As a result, the incorporation of published data continues to lag behind the pace of scientific publication, leaving large volumes of potentially valuable information inaccessible for computational analysis[11].

Compared with many scientific information extraction tasks, geoscientific knowledge is inherently organized around entities such as rock samples, stratigraphic units, fossil occurrences, drill cores, or geographic locations. These entities serve as the primary carriers of scientific observations, measurements, and interpretations, while the associated information is often distributed across multiple components of a publication[12,13]. Consequently, reconstructing a complete scientific record frequently requires integrating evidence from heterogeneous sources within a document rather than extracting information from isolated text passages[14,15]. Moreover, geoscientific records commonly exhibit hierarchical structures in which a single entity is associated with multiple attributes, measurements, analytical methods, uncertainties, and contextual metadata (e.g., GEOROC). Such information is rarely reported in a uniform format and may span different levels of granularity. These characteristics make direct document-level or sentence-level extraction insufficient for accurately recovering structured scientific knowledge. Instead, effective extraction requires an entity-centered process that can first identify target entities, then iteratively retrieve and assemble their associated attributes from distributed evidence sources. This challenge highlights the need for extraction frameworks capable of reconstructing hierarchical scientific records rather than merely recognizing isolated entities or relations.

Recent advances in large language models (LLMs) have expanded the capabilities of automated scientific information extraction. LLM-based systems have demonstrated strong capabilities in tasks such as entity recognition, relation extraction, information synthesis, and structured data generation[16–25]. In parallel, retrieval-augmented generation (RAG) and iterative retrieval methods have shown promise for accessing distributed evidence across long documents and supporting multi-step reasoning[26–35]. Despite these advances, most existing approaches have been developed for question answering, document understanding, or general-purpose information retrieval. Their ability to reconstruct complex, entity-centered scientific records from heterogeneous geoscientific literature remains largely unexplored. In particular, approaches capable of generalizing across multiple Earth science subdisciplines while producing structured, traceable, and verifiable outputs remain limited[36].

Recent studies have further emphasized the importance of multimodal data and tool-augmented reasoning in geoscience and remote sensing. OpenEarthMap-SAR and Bright provide large-scale benchmark datasets based on high-resolution SAR and multimodal optical–SAR observations, respectively, while GeoMMBench and GeoMMAgent demonstrate the potential of multimodal benchmarks and retrieval–perception–reasoning agents for geoscientific analysis[37–39]. Nevertheless, the reconstruction of structured, entity-centered records from heterogeneous scientific literature remains insufficiently explored.

Here we present GeoExtractor, a modular multi-agent framework for automated extraction of structured knowledge from geoscience literature. GeoExtractor formulates information extraction as a hierarchical, entity-centered reconstruction process in which entities are first identified and then linked to their associated attributes through structured extraction. The framework integrates an iterative decision–retrieval–generation workflow together with a two-stage retrieval mechanism to enable robust recovery of complex scientific records from heterogeneous document sources. We evaluate GeoExtractor using a benchmark spanning four geoscience subdisciplines (paleomagnetism, petroleum geology, geochemistry, and fire history), and compare its performance with representative extraction baselines. As a large-scale application, we further use GeoExtractor to construct a

CAOB database containing zircon U–Pb–Hf–O and whole-rock Sr–Nd isotope data, demonstrating its potential for accelerating geoscientific data synthesis and knowledge discovery.

The methodological design of GeoExtractor is characterized by three integrated elements: hierarchical entity-centered extraction, completeness-aware two-stage retrieval, and multi-model ablation-based evaluation. The framework decomposes record reconstruction into primary-key identification and field-level secondary-key extraction, thereby preserving the linkage between entities and their attributes. Its adaptive retrieval strategy uses extraction completeness to determine whether routine coverage-oriented retrieval is sufficient or whether progressive multi-hop retrieval is needed to integrate evidence across document sections. Finally, the multi-dataset, multi-model benchmark with repeated runs and component-level ablation evaluates the respective benefits and limitations of progressive retrieval and verification under different record structures rather than assuming that these components are uniformly positive.

Results

The GeoExtractor framework

GeoExtractor adopts a modular multi-agent architecture composed of six functional modules: document processing, primary key extraction, decision, retrieval, generation, and verification. The framework is built upon three key design principles. First, extraction is organized hierarchically. Rather than directly extracting all schema fields simultaneously, GeoExtractor first identifies primary-key instances that define individual records and then extracts the associated secondary-key fields for each instance. This decomposition transforms a complex record-level extraction task into a sequence of smaller and more manageable operations, reducing interference among heterogeneous fields and improving extraction stability in long scientific documents. Second, secondary-key extraction is performed through an iterative decision–retrieval–generation loop. For each record, the decision module determines the next field to be extracted, the retrieval module gathers the most relevant evidence from the document, and the generation module produces the field value. The process is repeated until all required secondary-key fields have been completed. This field-level iterative strategy enables the framework to dynamically adapt to varying document structures and information distributions while maintaining consistent extraction quality across large schemas. Third, the retrieval module employs a two-stage retrieval mechanism. The first stage is designed to handle the majority of routine extraction scenarios through efficient semantic retrieval of locally relevant evidence. When the retrieved context is insufficient or the target field requires information distributed across multiple document sections, a second-stage retrieval process is activated to perform multi-hop evidence gathering and reasoning. This hierarchical retrieval strategy balances efficiency and coverage, allowing GeoExtractor to address both straightforward extraction tasks and more challenging cases that require cross-paragraph information integration.

Task definition. Geoscientific data are typically organized around domain-specific entities, each associated with a set of constrained attributes. Although research objects and terminology vary substantially across subdisciplines, geoscientific datasets generally share a hierarchical entity–attribute structure. Given a geoscientific article together with user-defined field definitions, field descriptions, and parent–child relationships between fields, GeoExtractor produces structured records in which attribute values are explicitly linked to their corresponding entities. In the formal output schema, each entity is represented by a primary-key instance that serves as the unique identifier of an individual record. We use *entity* for the scientific object in the narrative and reserve *primary key* for its schema-level identifier and the corresponding extraction module or metric. The framework then extracts the associated secondary-key fields and establishes explicit links between field values and their parent primary-key instances. This formulation aligns with the underlying structure of geoscientific datasets. For example, in the GEOROC database, a rock sample is the entity represented by the primary-key sample identifier, whereas fields such as geographic location, Lu concentration, and Hf concentration correspond to secondary-key

fields. GeoExtractor must therefore identify individual sample entities while simultaneously assigning the correct secondary-key field values to each sample.

Benchmark evaluation

Benchmark design and construction

We constructed a cross-disciplinary geoscience information extraction benchmark to systematically evaluate the generalization capability of automated extraction methods across diverse sub-disciplines and heterogeneous data conditions. The benchmark comprises four datasets representing paleomagnetism, petroleum geology, geochemistry, and fire history, thereby covering a broad range of geoscientific data types and reporting styles.

Table 1 summarizes the benchmark statistics. The benchmark contains 150 publications, 2,980 entity records, and 28,104 attribute values. The datasets vary substantially in scale and structure: PaleoMag contains 896 records across 11 fields from 30 articles, whereas GCD contains 10 records across 8 fields from 10 articles. This variation reflects the diversity of real-world extraction scenarios, ranging from publications containing hundreds of records to studies reporting only a few observations.

The PaleoMag and SweetSpot datasets are derived from expert-curated compilations in paleomagnetism and petroleum exploration, respectively, and include high-quality ground-truth annotations verified by domain specialists. The GEOROC dataset was constructed by extracting a subset of records from the publicly available GEOROC Expert Datasets[40] and covers igneous rock samples with major and trace element compositions. The GCD dataset was derived from the Global Charcoal Database[41] and represents a challenging test case where each article corresponds to exactly one record, testing the method's robustness under low sample density conditions.

The benchmark encompasses publications with varying levels of structural and semantic complexity. Document lengths range from several to dozens of pages, and the number of fields to be extracted ranges from 8 to 16. In terms of modality, the benchmark includes unstructured text, structured and semi-structured tables, and figure captions, requiring methods to handle both text-based contextual reasoning and complex table understanding tasks such as large multi-page tables, hierarchical and cross-column headers, and semantic alignment between table bodies and annotation text.

Ground-truth validation and matching criteria

Ground-truth construction followed a source-tracing and expert-consistency procedure adapted from the consistency-validation strategy of Guo et al[13].. For PaleoMag and SweetSpot, records were checked against the original publications and expert-curated compilations. During validation, annotators verified not only individual field values but also the associations between each entity and its attributes, together with the relevant geographic, stratigraphic, geological, and analytical context. Values inconsistent with the source evidence or contextual relationships were corrected or excluded.

For evaluation, the ground-truth values were retained as close as possible to their original forms in the source datasets. Field-specific normalization was applied only when required for comparison. Geological ages were converted to Ma, and latitude and longitude values were converted to decimal degrees. For other numeric fields, the numerical values were compared directly with an absolute tolerance. When a prediction contained an uncertainty

Table 1 Benchmark dataset summary statistics.

Dataset	Publications	Entity Records	Attribute Types	Attribute Values
PaleoMag	30	896	11	9,856
SweetSpot	10	197	16	3,152
GEOROC	100	1,877	8	15,016
GCD	10	10	8	80
Total	150	2,980	43	28,104

expression such as ± or +/−, the numerical components in the prediction were compared with the numerical components in the reference value. Textual values were converted to lowercase and normalized by removing whitespace and punctuation. Primary keys were first matched after this normalization. A predicted secondary-key field was counted as correct only when it was associated with the corresponding matched primary-key instance and satisfied the relevant field-level matching rule. These matching rules were applied consistently across the benchmark datasets. The final labels were consolidated through expert review.

Extraction performance

We evaluated GeoExtractor against two baseline methods: (1) a full-text end-to-end generation approach that prompts an LLM to extract all fields in a single pass from the entire document, and (2) LangExtract[42], an open-source, example-driven information extraction framework that relies on user-provided examples for in-context learning. Each method was evaluated with DeepSeek-v4-flash, corresponding to the official DeepSeek-v4-flash-0731 release, together with Gemini-3.7-flash and GPT-5.4, using identical field definitions and prompt specifications. Each model–method–dataset configuration was run independently three times. Results are reported as the mean ± standard deviation across runs.

Table 2 reports recall, precision, and F1 as mean ± standard deviation across three independent runs, together with average total token consumption and processing time per document. GeoExtractor generally consumes more tokens and requires more processing time than the full-text baseline because its iterative decision–retrieval–generation workflow involves multiple LLM calls. Averaged across the three models, GeoExtractor obtained higher F1 than full-text generation on PaleoMag (0.9592 versus 0.9429) and SweetSpot (0.7821 versus 0.7427). On GEOROC, the GeoExtractor configurations achieved F1 scores of 0.4938, 0.4648, and 0.4602, compared with 0.4876, 0.4425, and 0.4718 for full-text generation across the three models. On average, the GeoExtractor configurations achieved a slightly higher F1 than full-text generation (0.4729 versus 0.4673), although the direction and magnitude of the difference depended on the underlying model. On GCD, GeoExtractor remained lower than full-text generation on average (0.6748 versus 0.7069). GeoExtractor outperformed LangExtract on all four datasets. The results therefore support the framework's advantage mainly for multi-record and multi-field extraction, while also confirming its limitation on single-record-per-article data. The two-stage progressive retrieval mechanism is intended to support multi-hop reasoning, such as GEOROC's `province` field, which requires linking `sample` → `sample location` → `province`. The ablation results in Table 3 further show its largest dataset-level gain on GEOROC.

Model-specific comparisons revealed heterogeneous effects. For GPT-5.4, GeoExtractor improved F1 relative to full-text generation on PaleoMag (0.9534 versus 0.9292), SweetSpot (0.7815 versus 0.7105), and GEOROC (0.4648 versus 0.4425), but underperformed on GCD (0.6609 versus 0.7181). DeepSeek-v4-flash showed a similar pattern, with gains on PaleoMag, SweetSpot, and GEOROC, but lower F1 on GCD (0.6395 versus 0.6961). Under Gemini-3.7-flash, the two methods were closer on the multi-record datasets and the direction of the difference was inconsistent: GeoExtractor was slightly lower on PaleoMag and GEOROC but higher on SweetSpot. It nevertheless achieved a modestly higher F1 on GCD (0.7240 versus 0.7065). These results suggest that the benefit of iterative retrieval depends both on the structure of the extraction task and on the underlying LLM. LangExtract performed substantially worse than both methods across all three models and datasets, although its relative performance varied by model. Because each configuration was repeated only three times, these model-specific differences are interpreted descriptively rather than as evidence of statistical significance.

Error analysis

To understand the nature and sources of extraction failures, we conducted a detailed field-level error analysis on the final GeoExtractor outputs. For each non-primary-key field with a non-empty reference value, we classified the result as correct, an omission when no value was extracted, or a wrong extraction when a value was produced

Table 2 Extraction performance and resource usage across the benchmark. Recall, precision, and F1 are reported as mean ± standard deviation across three independent runs. Average total token consumption and processing time per document are also reported.

Dataset	Model	Method	Recall	Precision	F1	Avg. Total Tokens/Doc. (K)	Avg. Time/ Doc. (s)
PaleoMag	DeepSeek-v4-flash	Full-text	0.9247 ± 0.0013	0.9429 ± 0.0165	0.9337 ± 0.0082	25	120
		GeoExtractor	0.9809 ± 0.0017	0.9405 ± 0.0040	0.9602 ± 0.0013	121	600
		LangExtract	0.7395 ± 0.0050	0.7904 ± 0.0019	0.7641 ± 0.0018	–	500
	GPT-5.4	Full-text	0.9110 ± 0.0950	0.9550 ± 0.0124	0.9292 ± 0.0464	20	60
		GeoExtractor	0.9743 ± 0.0017	0.9333 ± 0.0007	0.9534 ± 0.0010	85	285
		LangExtract	0.6968 ± 0.0377	0.8246 ± 0.0323	0.7537 ± 0.0086	–	300
	Gemini-3.7-flash	Full-text	0.9600 ± 0.0025	0.9715 ± 0.0006	0.9657 ± 0.0013	27	200
		GeoExtractor	0.9657 ± 0.0017	0.9620 ± 0.0070	0.9639 ± 0.0043	101	1230
		LangExtract	0.7943 ± 0.0598	0.8572 ± 0.0649	0.8246 ± 0.0623	–	900
SweetSpot	DeepSeek-v4-flash	Full-text	0.7732 ± 0.0179	0.7741 ± 0.0047	0.7736 ± 0.0106	39	220
		GeoExtractor	0.8138 ± 0.0195	0.7848 ± 0.0153	0.7989 ± 0.0148	171	800
		LangExtract	0.3013 ± 0.0213	0.7708 ± 0.0057	0.4328 ± 0.0229	–	480
	GPT-5.4	Full-text	0.6796 ± 0.0379	0.7456 ± 0.0045	0.7105 ± 0.0189	24	70
		GeoExtractor	0.8094 ± 0.0194	0.7556 ± 0.0079	0.7815 ± 0.0132	119	220
		LangExtract	0.3154 ± 0.0073	0.6962 ± 0.0702	0.4326 ± 0.0068	–	330
	Gemini-3.7-flash	Full-text	0.7313 ± 0.0008	0.7572 ± 0.0068	0.7441 ± 0.0031	27	225
		GeoExtractor	0.7497 ± 0.0042	0.7837 ± 0.0366	0.7660 ± 0.0200	141	555
		LangExtract	0.3728 ± 0.0194	0.7468 ± 0.0343	0.4963 ± 0.0097	–	3230
GEOROC	DeepSeek-v4-flash	Full-text	0.3860 ± 0.0039	0.6618 ± 0.0029	0.4876 ± 0.0023	53	239
		GeoExtractor	0.4014 ± 0.0197	0.6426 ± 0.0034	0.4938 ± 0.0142	239	561
		LangExtract	0.0942 ± 0.0011	0.5506 ± 0.0005	0.1608 ± 0.0015	–	264
	GPT-5.4	Full-text	0.3405 ± 0.0068	0.6322 ± 0.0028	0.4425 ± 0.0051	33	56
		GeoExtractor	0.3806 ± 0.0161	0.6001 ± 0.0295	0.4648 ± 0.0032	211	470
		LangExtract	0.1285 ± 0.0057	0.4775 ± 0.0387	0.2021 ± 0.0035	–	177
	Gemini-3.7-flash	Full-text	0.3674 ± 0.0211	0.6592 ± 0.0169	0.4718 ± 0.0217	46	15
		GeoExtractor	0.3662 ± 0.0117	0.6196 ± 0.0054	0.4602 ± 0.0077	266	480
		LangExtract	0.1535 ± 0.0186	0.4587 ± 0.0487	0.2301 ± 0.0271	–	730
GCD	DeepSeek-v4-flash	Full-text	0.7522 ± 0.0055	0.6480 ± 0.0075	0.6961 ± 0.0023	20	80
		GeoExtractor	0.7956 ± 0.0227	0.5351 ± 0.0217	0.6395 ± 0.0184	94	585
		LangExtract	0.1822 ± 0.0063	0.1860 ± 0.0187	0.1838 ± 0.0113	–	230
	GPT-5.4	Full-text	0.7867 ± 0.0109	0.6605 ± 0.0046	0.7181 ± 0.0067	13	10
		GeoExtractor	0.8089 ± 0.0063	0.5600 ± 0.0455	0.6609 ± 0.0340	57	110
		LangExtract	0.1867 ± 0.0013	0.1647 ± 0.0125	0.1747 ± 0.0071	–	150
	Gemini-3.7-flash	Full-text	0.7700 ± 0.0058	0.6526 ± 0.0077	0.7065 ± 0.0068	14	105
		GeoExtractor	0.7867 ± 0.0001	0.6707 ± 0.0124	0.7240 ± 0.0072	48	365
		LangExtract	0.1971 ± 0.0104	0.1232 ± 0.0540	0.1428 ± 0.0390	–	2080

Table 3 Ablation study averaged across three underlying LLMs. F1 scores are reported for Stage 1 only, Stage 1 + Stage 2, and the final configuration with verification.

Configuration	PaleoMag	SweetSpot	GEOROC	GCD
Stage 1 only	0.9605	0.7752	0.4504	0.6747
Stage 1 + Stage 2	0.9594	0.7768	0.4723	0.6743
Final (Stage 1 + Stage 2 + Verification)	0.9592	0.7821	0.4729	0.6748

but did not match the reference. In GEOROC, omissions accounted for 34,375 of 44,430 field errors (77.4%), while wrong extractions accounted for the remaining 10,055 errors. Thus, the dominant GEOROC failure mode is insufficient recovery of values that are present in the source, although the relative contribution of wrong extractions varies substantially by field, (Fig. 1).

The field-level pattern provides more specific insight into this behavior. Numerical geographic coordinates and age-related fields were particularly omission-prone: omissions constituted 83.3% of errors for `latitude`, 86.5% for `longitude`, 73.6% for `age`, and 96.2% for `uncertainty`. The isotope-ratio fields `yb` and `lu` showed a similar pattern, with omissions accounting for 92.7% and 94.9% of their respective errors. These fields may be reported in tables, captions, or heterogeneous textual formats, and values such as uncertainty are often separated from the corresponding measurement, making them difficult to recover consistently. In contrast, `province` errors were dominated by wrong extractions: 3,686 wrong values versus 1,889 omissions. This suggests that the model often identifies a geographically related value but confuses the hierarchical roles of province, larger region, and sample location. The results therefore indicate that GEOROC errors arise from both evidence-recovery failures for dispersed or irregularly formatted numeric fields and semantic disambiguation failures among closely related geographic fields.

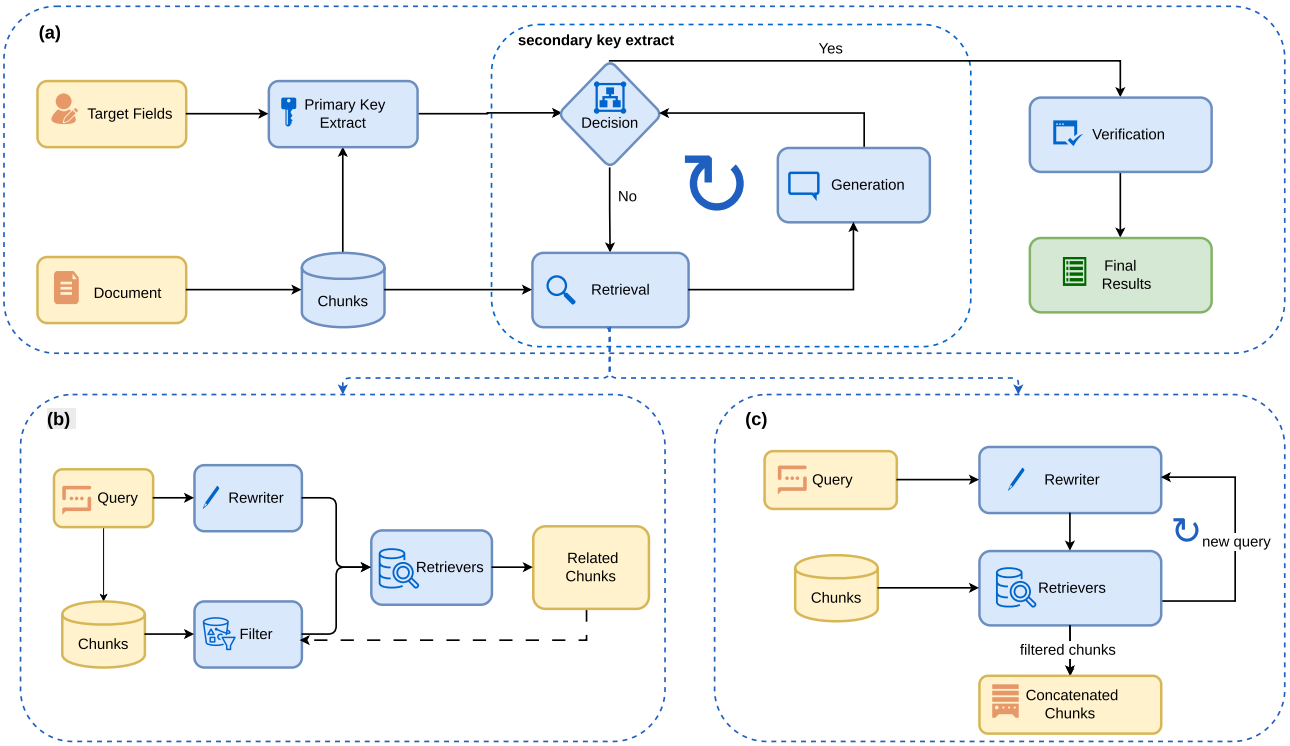


Fig. 1 Architecture of the GeoExtractor framework. (**a**) Overall modular agent architecture comprising six functional modules: document processing, primary key extraction, decision, retrieval, generation and verification. The decision module dynamically evaluates extraction completeness and controls iterative flow. (**b**) Stage 1 retrieval: coverage-driven iterative retrieval with marked-chunk filtering to ensure non-redundant full-text coverage across iterations. (**c**) Stage 2 retrieval: progressive multi-hop retrieval for complex fields requiring cross-paragraph reasoning, with LLM-driven query generation and important chunk extraction.

Application: data compilation for the Central Asian orogenic belt

Database construction and tectonic background

To evaluate the practical utility of GeoExtractor for large-scale database construction, we selected the Central Asian Orogenic Belt (CAOB) as an application domain because it provides a data-rich and heterogeneous test bed for scientific information extraction. The CAOB, also known as the Altaids, is a large accretionary orogen extending from the Ural Mountains to the Pacific margin and comprising continental fragments, magmatic arcs, and subduction–accretion complexes[43–53].

We applied the framework to 179 CAOB-related publications retrieved from the Web of Science. The resulting database contains 2,259 zircon analyses, including U–Pb ages, Hf–O isotopes, whole-rock Sr–Nd isotopes, and associated sample metadata. The framework processed each publication in approximately 15 minutes, compared with approximately 8 hours for our manual extraction workflow. The 8-hour value is our own empirical estimate based on manual extraction experience and is broadly comparable in magnitude to the more than 10 hours that Guo et al. reported that students needed, on average, to manually collect 100 data entries[13]. For the 179-publication corpus, this corresponds to approximately 1,432 hours of manual work and 44.75 hours of automated processing, yielding a ratio of approximately 32:1, which we report as an approximately 30-fold reduction in processing time.

Manual audit of the CAOB database

Agreement with previously reported tectonic interpretations provides only a plausibility check, not record-level accuracy. We therefore traced records from two CAOB publications to their original tables and text. The audited U–Pb publication yielded an all-key recall of 78.05%, precision of 94.49%, and F1 score of 85.49%, whereas the Sr–Nd publication yielded 30.95%, 16.88%, and 21.85%, respectively. U–Pb errors mainly involved incomplete recovery and confusion among age columns; Sr–Nd extraction additionally showed record over-generation and misalignment among isotope ratios, model ages, and sample-level versus zircon-spot ages.

This heterogeneity is also evident in the 100-article GEOROC results, where document-level field F1 ranged from 0 to 0.875; five articles scored below 0.1 and ten above 0.7. The CAOB database should therefore be treated as a literature-derived compilation requiring human quality control, rather than as a uniformly complete or error-free resource.

We also quantified field-level availability in the released database (Table 4). U–Pb age, rock type, and location were available for 99.2%, 99.9%, and 92.3% of U–Pb records, respectively, whereas latitude and longitude were available for 36.9%. For Hf–O records, $^{176}Hf/^{177}Hf$, $\varepsilon Hf(t)$, and crustal TDM were available for 88.4%, 82.7%, and 85.5%, respectively, while $\delta^{18}O$ was available for 15.3%. These are observed non-empty rates rather than completeness estimates, because a field may be absent from the source publication. A statistically representative completeness estimate for all 2,259 records would require a larger source-level audit.

Temporal distribution of magmatism and crustal age structure

The zircon U–Pb age distribution (Fig. 2) shows a dominant peak at 500–300 Ma, a secondary concentration at 1000–800 Ma, and sparse records at 2000–1500 Ma. Whole-rock Nd model ages are mainly concentrated at 1.2–0.8 Ga, whereas zircon Hf model ages extend into the older Precambrian, with tails reaching 3–4 Ga. Together, these distributions indicate that Paleozoic magmatism involved predominantly juvenile accreted crust as well as subordinate contributions from older crustal components[54,55].

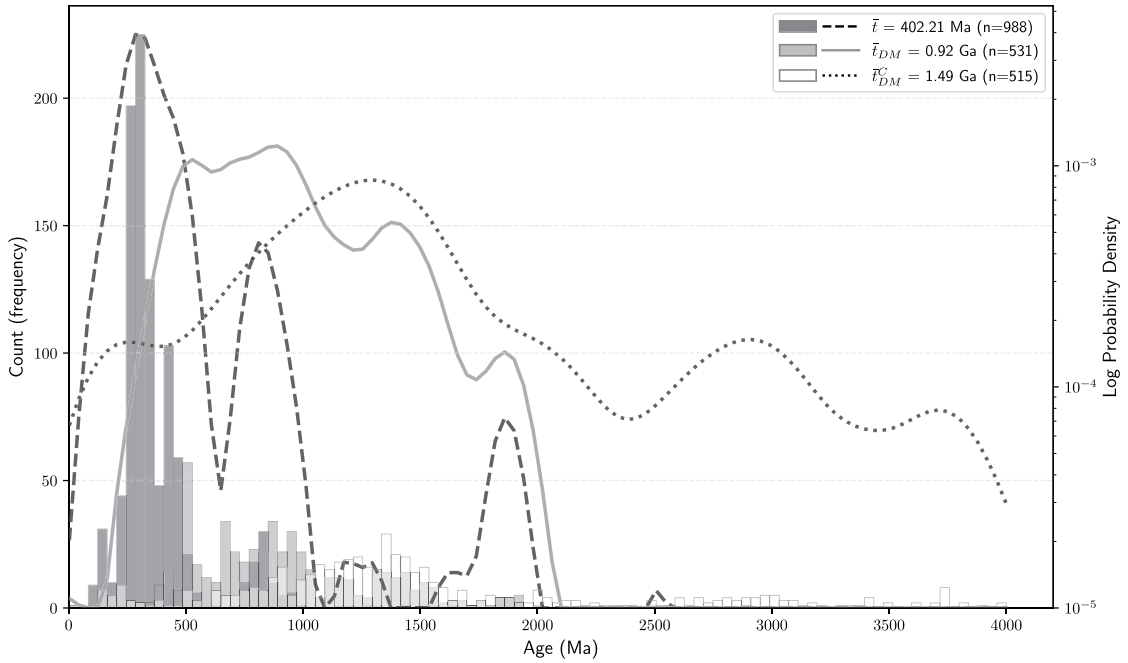


Fig. 2 Histograms and probability density curves for the CAOB magmatic rocks. (**a**) Zircon U–Pb ages, showing a dominant peak at 500–300 Ma, a secondary peak at 1000–800 Ma, and sparse ages at 2000–1500 Ma. (**b**) Zircon Hf model ages, exhibiting Precambrian tails extending to 3–4 Ga. (**c**) Whole-rock Nd model ages, concentrated at 1.2–0.8 Ga with minimal Archean contributions.

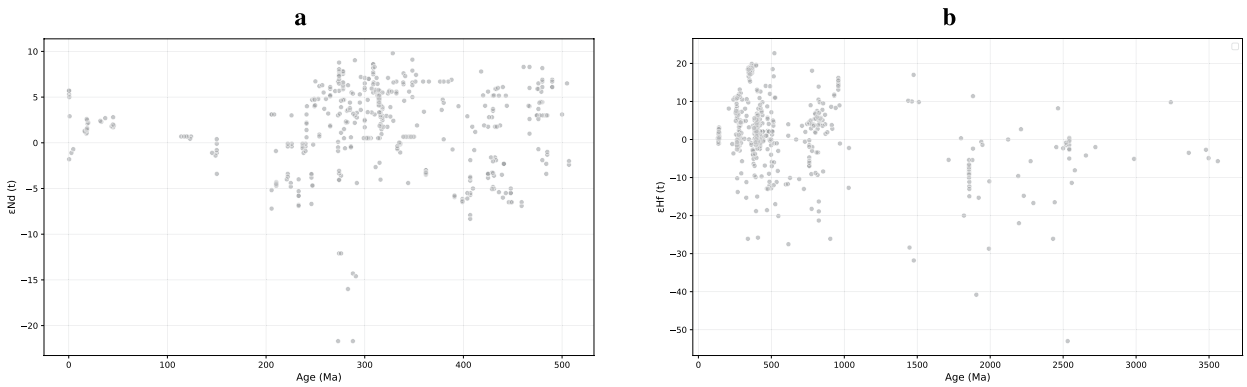


Fig. 3 Integrated Sr–Nd and zircon Hf–O isotopic data for the Central Asian Orogenic Belt. (**a**) Sr–Nd isotopic data for igneous rocks, including $\varepsilon Nd(t)$ values from −22 to +8 and initial $^{87}Sr/^{86}Sr$ ratios mainly between 0.703 and 0.707. (**b**) zircon Hf–O isotopic data, showing positive, intermediate, and negative $\varepsilon Hf(t)$ populations and $\delta^{18}O$ values concentrated at 5.5–8.5‰.

Source mixing constrained by Sr–Nd isotopes

The Sr–Nd data (Fig. 3a) show a broad range of $\varepsilon Nd(t)$ values from +8 to −22, with most samples clustering between +3 and +9, while initial $^{87}Sr/^{86}Sr$ ratios are mainly concentrated at 0.703–0.707. These patterns are consistent with mixed juvenile and evolved crustal sources and do not show a strong secular trend in crustal recycling.

Crust–mantle interaction revealed by zircon Hf–O isotopes

Zircon $\varepsilon Hf(t)$ values (Fig. 3b) form positive, intermediate, and negative populations, with the most negative values mainly associated with older apparent ages. Zircon $\delta^{18}O$ values are concentrated at 5.5–8.5‰, slightly above the mantle zircon baseline, and show a plateau around 400–500 Ma. These distributions indicate mixing between juvenile and ancient crustal components during CAOB magmatism.

Integrated understanding of CAOB crustal evolution

Taken together, the extracted isotopic data are consistent with a long-lived, multi-stage subduction-accretion history of the CAOB, involving substantial juvenile crust production, subsequent reworking, and subordinate incorporation of ancient crustal material[46,49,52,56–58].

This case study demonstrates how automated literature extraction can support regional-scale geoscientific synthesis.

Discussion

This study demonstrates that a modular LLM-based agent framework can effectively extract structured quantitative information from geoscientific literature without task-specific fine-tuning or extensive rule engineering. The effectiveness of GeoExtractor primarily arises from decomposing document-level extraction into hierarchical sub-problems organized around primary-key instances and secondary-key fields, which allows the model to localize reasoning at the level of individual fields rather than processing entire documents in a single pass. This design, combined with an iterative decision–retrieval–generation process and a two-stage retrieval mechanism, enables adaptive allocation of computational effort, with simple fields resolved through lightweight retrieval and complex fields triggering deeper multi-hop evidence aggregation.

Compared with end-to-end generation approaches and example-dependent systems such as LangExtract, GeoExtractor reduces reliance on global document understanding and curated demonstrations, instead leveraging field-driven control to generalize across geoscience sub-disciplines. The two-stage retrieval strategy further contributes to robustness by separating efficient coverage-oriented retrieval from selective reasoning-oriented retrieval, ensuring that computational resources are focused where they are most needed.

From a practical perspective, GeoExtractor substantially reduces the time required for literature-based data compilation, processing each article in approximately 15 minutes including retrieval and generation steps. While manual extraction remains more flexible in handling ambiguous cases, it typically requires several hours per article. The token and wall-clock measurements in Table 2 make the computational trade-off explicit: the iterative framework generally requires more LLM resources than a single-pass full-text method. The reported processing times are empirical averages under the experimental conditions and may vary with API load and time of day. These results suggest that automated extraction frameworks can significantly improve the scalability of geoscientific data synthesis. The most appropriate use of GeoExtractor is therefore as a large-scale preliminary compilation and review-prioritization system: the machine supplies speed and broad coverage, whereas domain experts remain responsible for final accuracy and quality control[59,60].

Several limitations should be acknowledged. First, on datasets where each article contains only a single record (GCD), GeoExtractor sometimes over-generates candidate records, and the simple full-text baseline remains better on average across the three LLMs (mean F1 0.7069 vs. 0.6748). GCD differs qualitatively from the other datasets because each article contributes only one target entity. For GCD, the primary entity is the site name, defined as the "Location of historical fire recorded in this study." GeoExtractor may extract several location mentions from different text chunks as separate candidate entities, even when a location is not associated with a historical fire reported in the present study. It may also extract locations mentioned only in cited studies. These predictions

are grounded in text spans from the article, but violate the target entity constraint because they lack the required historical-fire association or refer to external studies. They are therefore counted as false positives, producing the observed precision collapse. The full-text baseline is less affected because it processes the article globally in a single pass and is less likely to interpret separate chunks as separate entities. More broadly, this result demonstrates that GeoExtractor's advantages are most pronounced in multi-record scenarios—which constitute the majority of practical geoscience extraction tasks—and that alternative approaches may be preferable for single-record-per-article datasets. Second, although the benchmark has been expanded to 150 publications, the number of stochastic repetitions remains small at three runs per configuration. Third, the current benchmark covers four sub-disciplines of geoscience; validation on a broader range of domains, including sedimentology, hydrology, and atmospheric science, would further establish cross-disciplinary generalizability. Fourth, the ground-truth annotations for the benchmark were produced by a single annotator; inter-annotator agreement rates are therefore not available, and the annotation quality ceiling is unknown. Establishing multi-annotator ground truth with reported agreement metrics would strengthen the reliability of the benchmark. Fifth, the current framework processes text and tables but does not natively interpret figures, maps, or stratigraphic columns—modalities that contain substantial information in geoscience publications. Integrating vision-language models for multi-modal extraction represents an important direction for future work. Additionally, standard isotope geochemistry quality-control concerns—such as discordance filtering for zircon U–Pb ages, harmonization of Hf model age calculation parameters, and treatment of inter-laboratory calibration offsets—are not addressed in the current automated pipeline and represent important considerations for production use. Sixth, the LangExtract baseline comparison employs the framework in its simple-example configuration; results with domain-specific examples may differ. Finally, the extraction quality depends on the capabilities of the underlying LLM; improvements in base model reasoning, context length, and factual accuracy will directly benefit the framework.

Future work should pursue several directions. Integrating figure and table understanding capabilities, potentially through vision-language models, would enable true multi-modal extraction from geoscience publications. Incorporating domain-specific knowledge graphs could improve entity linking and cross-document data harmonization. The present study supplied relatively general field definitions but did not provide extensive domain-specific background knowledge, analytical conventions, or field-specific examples. Providing such prior knowledge to the decision, retrieval, generation, and verification modules could help distinguish relevant from irrelevant values and reduce omission and primary-key–secondary-key misalignment errors. Scaling the framework to operate on continuous publication streams through automatically extracting and updating databases as new papers appear, could enable living geoscience knowledge bases that grow with the literature. Extending the approach to related data-rich disciplines such as environmental science, soil science, and oceanography would further demonstrate the generality of the iterative agent extraction paradigm.

In summary, GeoExtractor establishes a practical methodology for converting the vast, unstructured knowledge embedded in geoscience literature into structured, queryable databases suitable for computational analysis. By enabling systematic, scalable literature data synthesis, this approach addresses a critical bottleneck in data-driven Earth system science and provides a foundation for next-generation geoscience knowledge infrastructure.

Methods

Evaluation metrics

To quantitatively evaluate extraction performance, we employ precision, recall, and F1 score at the field level. The evaluation is conducted by comparing model outputs against human-annotated ground truth for each field value:

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}}$$
$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}$$
$$\text{F1} = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$$

where TP (true positive) denotes correctly extracted field values, FP (false positive) denotes spuriously extracted values (hallucinated or incorrect), and FN (false negative) denotes missed field values that exist in the ground truth but were not extracted by the model. This field-level metric balances omission errors (recall) and hallucination errors (precision), providing a comprehensive measure of joint primary-key–secondary-key extraction capability.

GeoExtractor architecture

GeoExtractor is implemented as a modular multi-agent system. All LLM-driven modules use a selected LLM accessed via API. The framework is model-agnostic; in the benchmark we evaluated it with DeepSeek-v4-flash, corresponding to the official DeepSeek-v4-flash-0731 release, together with Gemini-3.7-flash and GPT-5.4.

Document processing. Input articles are first parsed from PDF into markdown format using MinerU2.5[61]. For tables, the table regions identified during PDF parsing are further converted into structured LaTeX representations using Table2LaTeX-RL[62]. The markdown text is separated into text and table components. Text is segmented into chunks of 1,024 characters using a sliding window with no overlap. Tables are preserved as whole chunks regardless of size. All chunks are indexed and stored in a chunk library for retrieval. Published component-level evaluations provide reference points for the fidelity of this conversion pipeline. MinerU2.5 reports an overall score of 90.67 on OmniDocBench, a text edit distance of 0.047, a table TEDS score of 88.22, and a reading-order edit distance of 0.044[61]. For complex tables, Table2LaTeX-RL reports a compile ratio of 0.9917, a TEDS score of 0.8673, and a TEDS-Structure score of 0.9218[62].

Primary key extraction. The primary key extraction module processes the full article text in a single LLM call. The prompt instructs the LLM to identify all instances of the primary key, outputting each instance with a unique identifier. For example, for a geochemistry dataset, the LLM identifies sample IDs. The extracted primary keys serve as anchors for subsequent secondary-key extraction.

Decision module. After primary-key extraction initializes the extraction state, Stage 1 is triggered whenever the target secondary-key fields remain incomplete. At each iteration, the decision module receives: (1) the current cumulative extraction results, (2) the list of target secondary-key fields with their completion status, and (3) the set of remaining unprocessed chunks. It evaluates whether all target fields are sufficiently covered. If fields remain incomplete, it generates a natural-language query describing the information still needed, which is passed to the retrieval module. Stage 1 terminates when the decision module indicates completion, when a generation round produces no new incremental field values, when all available chunks have been processed, or when the remaining workflow steps reach the configured safety threshold. In the reported configuration, the no-progress threshold was one consecutive round and the remaining-step threshold was five.

Retrieval module. The retrieval module combines sparse retrieval (BM25) and dense retrieval (text embeddings from a sentence transformer model). The query from the decision module is first rewritten by an LLM-based query rewriting model to optimize semantic coverage. The rewritten query is then issued to both retrievers in parallel. Retrieved results are merged by taking the union of the top-k results from each retriever ($k = 10$ for BM25, $k = 10$ for dense), with duplicates removed.

In Stage 1, all chunks retrieved and passed to the generation module are marked. Marked chunks are excluded from subsequent retrieval rounds (implemented via an ID-based filter on the retrieval results). This ensures that each retrieval round targets only previously unprocessed chunks.

Stage 2 is entered after the Stage 1 result is finalized and first re-evaluates the accumulated extraction state. If the target fields remain incomplete, it performs iterative query rewriting and multi-hop retrieval for up to three rewriting iterations. In each iteration, the query and retrieved chunks are input to the LLM, which outputs: (a) important chunk indices within <important_ids> tags, identifying the most relevant chunks from the current retrieval set; (b) a refined query within <query> tags for the next iteration. The rewriting loop terminates earlier if the search is judged complete, if the accumulated selected chunks exceed the configured limit, or if no further chunks are available. The Stage 2 decision process also terminates when no new field values are produced, when the decision module indicates completion, or when the remaining-step safety threshold is reached. Important chunks from all Stage 2 iterations are accumulated and passed to the generation module. If Stage 1 has already completed the target fields, Stage 2 exits without additional retrieval.

Generation module. The generation module is a single LLM call that receives: (1) the query specifying which fields to extract, (2) the set of retrieved chunks, and (3) field definitions with descriptions. It outputs structured field values in JSON format. The generation is field-level: the LLM extracts only the specified target fields from the provided chunks, without access to historical extraction results or process state. After each generation round, results are merged with the cumulative extraction state by updating field values (new values added, conflicts resolved by keeping the most recently extracted value). The merged state is returned to the decision module.

Verification module. After the iterative cycle terminates, the final aggregated results and all marked text chunks are passed to the verification module. The LLM re-examines each extracted field value against the source chunks and provides feedback on whether the value has supporting evidence in the original text. This field-level evidence feedback improves source traceability and enables human auditors to identify values that require further review. In the final processing configuration, values lacking sufficient textual support may also be filtered from the output. Verification is therefore intended primarily as an evidence-feedback and audit-support step, rather than as a substitute for expert validation or an independent extraction stage.

Ablation analysis

To quantify the contribution of progressive retrieval and verification, we evaluated three configurations: Stage 1 only, Stage 1 + Stage 2, and the final configuration (Stage 1 + Stage 2 + Verification). The results in Table 3 are averaged across DeepSeek-v4-flash, corresponding to the official DeepSeek-v4-flash-0731 release, Gemini-3.7-flash, and GPT-5.4.

Stage 1 only provides coverage-driven iterative retrieval without progressive multi-hop retrieval or the final verification step. Adding Stage 2 (progressive multi-hop retrieval[63]) improved the mean F1 score on GEOROC from 0.4504 to 0.4723, an absolute gain of 0.0219 and the largest gain among the four datasets. This result is consistent with the role of progressive retrieval in recovering evidence distributed across multiple sections. The improvement was small on SweetSpot (0.7752 to 0.7768) and negligible on PaleoMag (0.9605 to 0.9594) and GCD (0.6747 to 0.6743), indicating that these datasets benefit less from additional multi-hop retrieval under the evaluated settings.

Adding the verification module also produced dataset-dependent effects. Its primary role is to provide field-level feedback on whether each extracted value is supported by evidence in the source text, thereby facilitating subsequent human review; changes in F1 are a secondary performance consequence of using this feedback for filtering. On SweetSpot, mean F1 increased from 0.7768 to 0.7821, while on GEOROC it increased slightly from 0.4723 to 0.4729. The effect was negligible on PaleoMag (0.9594 to 0.9592) and GCD (0.6743 to 0.6748). Overall, the ablation results indicate that verification should be understood primarily as an evidence-traceability and audit-support mechanism, whose effect on F1 depends on the structure and evidence distribution of the target dataset rather than improving performance uniformly.

Baseline methods

Two baseline methods were implemented for comparison using the same three underlying LLMs: DeepSeek-v4-flash, corresponding to the official DeepSeek-v4-flash-0731 release, Gemini-3.7-flash, and GPT-5.4. For every model–method–dataset configuration, we performed three independent runs with the same prompts, field definitions, and evaluation rules. Recall, precision, and F1 are reported as the mean ± standard deviation calculated across the three runs. Token usage was collected directly from the usage metadata returned by each LLM API call, rather than estimated from input text length. Token counts from all calls for a document were aggregated and reported as average total token consumption per document. Because LangExtract does not provide a direct interface for token-usage statistics, its token consumption was not reported in the present study. Processing time was measured as wall-clock time from the start to the completion of document processing and extraction, and is reported as an empirical average across runs.

Full-text generation. The entire article text is provided to the LLM in a single prompt, together with field definitions identical to those used by GeoExtractor. The LLM is instructed to extract all field values in a single pass. For articles exceeding the model's context window, the text is truncated to the maximum allowable length. The complete prompt is provided in Additional Information, Section AI.1.

LangExtract. We configured LangExtract[42], Google's open-source example-driven extraction framework, with the same field definitions. For each dataset, we provided two in-context examples: one representative text passage and one representative table, each paired with manually annotated extraction results. The examples were selected from the training/development portion of the dataset, and no evaluation document was used as a demonstration. During execution, each example was converted into LangExtract's native `ExampleData` structure containing an `Extraction` object, with the primary key represented by its verbatim source span and the remaining fields stored as attributes. The framework was used with its default generic extraction configuration for each of the three LLMs. Additional examples were tested but did not produce consistent improvements, so two complementary examples were used to balance structural coverage and prompt length. The prompt template and demonstration examples are provided in Additional Information, Section AI.2.

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Data availability The benchmark datasets and CAOB extracted database in this study have been deposited in Zenodo and are available at DOI https://doi.org/10.5281/zenodo.22854445.

Code availability The source code for GeoExtractor is publicly available at Zenodo: https://doi.org/10.5281/zenodo.22854543 .

Declarations

Competing interests The authors declare no competing interests.

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